

ELITE IGCSE ACADEMY

Private Past Paper Solutions

Jan 2022 P1H Worked Solutions

Jan 2022 | P1H

29 questions ■ question image followed by worked solution

PREPARED BY

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PAST PAPER QUESTION**Q#: 1****Marks: 6**TOPIC: **Algebraic Roots & Indices**

Jan 2022 P1H - Jan 2022 | P1H

1 (a) Simplify $a^7 \times a^4$
(1)(b) Simplify $w^{15} \div w^3$
(1)(c) Simplify $(8x^5y^3)^2$
(2)(d) Make t the subject of $c = t^3 - 8v$
(2)**(Total for Question 1 is 6 marks)**

PAST PAPER SOLUTION**Q#: 1****Marks: 6**TOPIC: **Algebraic Roots & Indices**

Jan 2022 P1H - Jan 2022 | P1H

WORKED SOLUTION*Dr. Eslam Ahmed***Topic check.** Corrected to algebraic roots and indices. Index laws carry most of the marks.**Solution**

(a)

$$a^7 \times a^4 = a^{11}$$

(b)

$$w^{15} \div w^3 = w^{12}$$

(c)

$$(8x^5y^3)^2 = 64x^{10}y^6$$

(d)

$$c = t^3 - 8v$$

$$t^3 = c + 8v$$

FINAL ANSWER

$$a^{11}, w^{12}, 64x^{10}y^6, t = \sqrt[3]{c + 8v}$$

PAST PAPER QUESTION**Q#: 2****Marks: 3****TOPIC: Ratio Problem Solving**

Jan 2022 P1H - Jan 2022 | P1H

- 2 Danil, Gabriel and Hadley share some money in the ratios 3:5:9
The difference between the amount of money that Gabriel receives and the amount of money that Hadley receives is 196 euros.

Work out the amount of money that Danil receives.

..... euros

(Total for Question 2 is 3 marks)

PAST PAPER SOLUTION**Q#: 2****Marks: 3****TOPIC: Ratio Problem Solving**

Jan 2022 P1H - Jan 2022 | P1H

WORKED SOLUTION*Dr. Eslam Ahmed*

Topic check. Corrected to Ratio Problem Solving. This is a ratio from a difference, not exchange rates.

Method

The ratio is

$$\text{Danil} : \text{Gabriel} : \text{Hadley} = 3 : 5 : 9$$

The difference between Gabriel and Hadley is

$$9 - 5 = 4 \text{ parts}$$

These 4 parts are 196 euros, so one part is

$$196 \div 4 = 49$$

Danil has 3 parts:

$$3 \times 49 = 147$$

FINAL ANSWER

Danil receives 147 euros.

PAST PAPER QUESTION**Q#: 3****Marks: 3****TOPIC: Right-Angled Triangles - Pythagoras & Trigonometry**

Jan 2022 P1H - Jan 2022 | P1H

- 3 The diagram shows triangle ABC

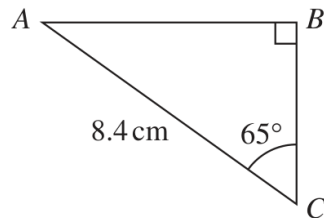


Diagram **NOT**
accurately drawn

Work out the length of the side AB
Give your answer correct to 3 significant figures.

..... cm

(Total for Question 3 is 3 marks)

PAST PAPER SOLUTION**Q#: 3****Marks: 3****TOPIC:** Right-Angled Triangles - Pythagoras & Trigonometry

Jan 2022 P1H - Jan 2022 | P1H

WORKED SOLUTION*Dr. Eslam Ahmed*

Topic check. Corrected to Right-Angled Triangles - Pythagoras & Trigonometry. This is a right-angled trigonometry problem.

Solution

Side AB is opposite the 65° angle, and $AC = 8.4$ cm is the hypotenuse.

$$\sin 65^\circ = \frac{AB}{8.4}$$

$$AB = 8.4 \sin 65^\circ = 7.613 \dots$$

FINAL ANSWER

7.61 cm.

PAST PAPER QUESTION**Q#: 4****Marks: 5****TOPIC: Percentages**

Jan 2022 P1H - Jan 2022 | P1H

- 4 Sarah makes and sells mugs.

One day she makes 150 mugs.

Her total cost for making these mugs is £1140

Of these mugs

$\frac{2}{5}$ are small mugs

32% are medium mugs

and the rest are large mugs

Here is Sarah's price list for selling each mug.

MUGS	
Small	£8.50
Medium	£11.20
Large	£14.20

Sarah sells all 150 mugs.

Work out her percentage profit.

Give your answer correct to the nearest whole number.

.....%

(Total for Question 4 is 5 marks)

PAST PAPER SOLUTION**Q#: 4****Marks: 5**TOPIC: **Percentages**

Jan 2022 P1H - Jan 2022 | P1H

WORKED SOLUTION*Dr. Eslam Ahmed***Topic check.** Percentages. The tag is correct.**Method**

Small mugs:

$$\frac{2}{5} \times 150 = 60$$

Medium mugs:

$$32\% \times 150 = 48$$

Large mugs:

$$150 - 60 - 48 = 42$$

Revenue:

$$\begin{aligned} &60(8.50) + 48(11.20) + 42(14.20) \\ &= 510 + 537.60 + 596.40 = 1644 \end{aligned}$$

Profit:

$$1644 - 1140 = 504$$

Percentage profit:

$$\frac{504}{1140} \times 100 = 44.21 \dots$$

Correct to the nearest whole number:

$$44\%$$

FINAL ANSWER

44%

PAST PAPER QUESTION**Q#: 5** **Marks: 6**TOPIC: **Statistics Toolkit**

Jan 2022 P1H - Jan 2022 | P1H

5 Jenny has six cards.

Each card has a whole number written on it so that

- the smallest number is 5
- the largest number is 24
- the median of the six numbers is 14
- the mode of the six numbers is 8

Jenny arranges her cards so that the numbers are in order of size.

5					24
---	-------	-------	-------	-------	----

- (a) For the remaining four cards, write on each dotted line a number that could be on the card.

(3)

A basketball team plays 6 games.

After playing 5 games, the team has a mean score of 21 points per game.

After playing 6 games, the team has a mean score of 23 points per game.

- (b) Work out the number of points the team scored in its 6th game.

.....
(3)**(Total for Question 5 is 6 marks)**

PAST PAPER SOLUTION**Q#: 5****Marks: 6****TOPIC: Statistics Toolkit**

Jan 2022 P1H - Jan 2022 | P1H

WORKED SOLUTION*Dr. Eslam Ahmed***Topic check.** Statistics Toolkit. The tag is correct.**Solution**

One possible ordered set is

$$5, 8, 8, 20, 21, 24$$

Check:

$$\text{median} = \frac{8 + 20}{2} = 14$$

The mode is 8, the smallest number is 5, and the largest number is 24.

For the basketball scores:

$$5 \times 21 = 105$$

$$6 \times 23 = 138$$

$$138 - 105 = 33$$

FINAL ANSWER

5, 8, 8, 20, 21, 24 and 33.

PAST PAPER QUESTION**Q#: 6****Marks: 5****TOPIC: Factorising**

Jan 2022 P1H - Jan 2022 | P1H

6 (a) Solve the inequality $5x - 7 \leq 2$

.....
(2)

(b) (i) Factorise $y^2 - 2y - 35$

.....
(2)

(ii) Hence, solve $y^2 - 2y - 35 = 0$

.....
(1)

(Total for Question 6 is 5 marks)

PAST PAPER SOLUTION**Q#: 6****Marks: 5****TOPIC: Factorising**

Jan 2022 P1H - Jan 2022 | P1H

WORKED SOLUTION*Dr. Eslam Ahmed***Topic check.** Factorising. The tag is correct.**Solution**

(a)

$$5x - 7 < 2$$

$$5x < 9$$

$$x < \frac{9}{5}$$

(b)(i)

$$y^2 - 2y - 35 = (y - 7)(y + 5)$$

(b)(ii) $y = 7$ or $y = -5$.**FINAL ANSWER**(a) $x < \frac{9}{5}$, (b)(i) $(y - 7)(y + 5)$, (b)(ii) $y = 7$ or $y = -5$

PAST PAPER QUESTION**Q#: 7****Marks: 3****TOPIC: Set Notation & Venn Diagrams**

Jan 2022 P1H - Jan 2022 | P1H

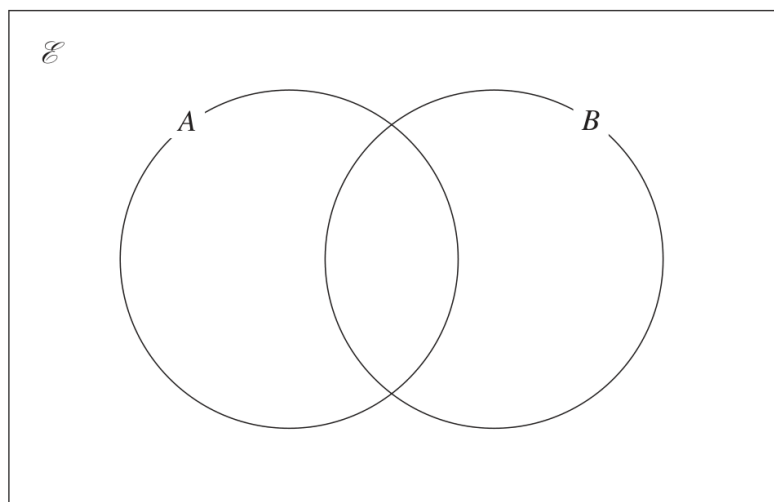
7 $\mathcal{E} = \{4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15\}$

$$A \cap B = \{5, 10, 15\}$$

$$B' = \{7, 8, 9, 11, 12, 13, 14\}$$

$$A' = \{4, 6, 7, 8, 14\}$$

Complete the Venn diagram for this information.



(Total for Question 7 is 3 marks)

PAST PAPER SOLUTION**Q#: 7****Marks: 3****TOPIC: Set Notation & Venn Diagrams**

Jan 2022 P1H - Jan 2022 | P1H

WORKED SOLUTION*Dr. Eslam Ahmed***Topic check.** Set notation and Venn diagrams. The tag is correct.**Method**

$$\mathcal{E} = \{4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15\}$$

We are given

$$A \cap B = \{5, 10, 15\}$$

Also,

$$B' = \{7, 8, 9, 11, 12, 13, 14\}$$

so

$$B = \{4, 5, 6, 10, 15\}$$

And

$$A' = \{4, 6, 7, 8, 14\}$$

so

$$A = \{5, 9, 10, 11, 12, 13, 15\}$$

Therefore the Venn diagram regions are:

- $A \cap B$: $\{5, 10, 15\}$
- A only: $\{9, 11, 12, 13\}$
- B only: $\{4, 6\}$
- outside both sets: $\{7, 8, 14\}$

FINAL ANSWERPut $\{5, 10, 15\}$ in the overlap, $\{9, 11, 12, 13\}$ in A only, $\{4, 6\}$ in B only, and $\{7, 8, 14\}$ outside.

PAST PAPER QUESTION**Q#: 8****Marks: 2****TOPIC: Powers, Roots & Standard Form**

Jan 2022 P1H - Jan 2022 | P1H

8

$$a = 4.2 \times 10^{-24}$$

$$b = 3 \times 10^{145}$$

Work out the value of $a \times b$

Give your answer in standard form.

(Total for Question 8 is 2 marks)

Dr. Eslam

PAST PAPER SOLUTION**Q#: 8****Marks: 2****TOPIC: Powers, Roots & Standard Form**

Jan 2022 P1H - Jan 2022 | P1H

WORKED SOLUTION*Dr. Eslam Ahmed***Topic check.** Powers, roots and standard form. The tag is correct.**Method**

$$a = 4.2 \times 10^{-24}$$

$$b = 3 \times 10^{145}$$

So

$$ab = (4.2 \times 3) \times 10^{-24+145}$$

$$ab = 12.6 \times 10^{121}$$

Write this in standard form:

$$12.6 \times 10^{121} = 1.26 \times 10^{122}$$

FINAL ANSWER

$$1.26 \times 10^{122}$$

PAST PAPER QUESTION**Q#: 9****Marks: 4****TOPIC: Sine, Cosine Rule & Area of Triangles**

Jan 2022 P1H - Jan 2022 | P1H

- 9 The diagram shows isosceles triangle ABC

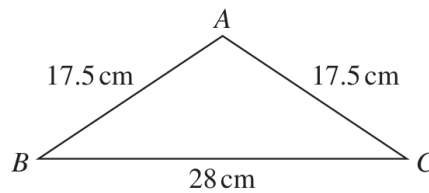


Diagram **NOT**
accurately drawn

$$AB = AC = 17.5 \text{ cm}$$

$$BC = 28 \text{ cm}$$

Calculate the area of triangle ABC

..... cm^2

(Total for Question 9 is 4 marks)

PAST PAPER SOLUTION**Q#: 9****Marks: 4****TOPIC: Sine, Cosine Rule & Area of Triangles**

Jan 2022 P1H - Jan 2022 | P1H

WORKED SOLUTION*Dr. Eslam Ahmed***Topic check.** Sine, Cosine Rule & Area of Triangles. The tag is correct.**Solution**

Split the isosceles triangle into two right-angled triangles.

$$\frac{BC}{2} = 14$$

$$h^2 = 17.5^2 - 14^2 = 110.25$$

$$h = 10.5$$

$$\text{area} = \frac{1}{2} \times 28 \times 10.5 = 147$$

FINAL ANSWER147 cm².

PAST PAPER QUESTION**Q#: 10****Marks: 3**TOPIC: **Coordinate Geometry**

Jan 2022 P1H - Jan 2022 | P1H

10 The straight line **L** has equation $2y + 7x = 10$

(a) Find the gradient of **L**

.....
(2)

(b) Find the coordinates of the point where **L** crosses the y-axis.

(..... ,)
(1)

(Total for Question 10 is 3 marks)

PAST PAPER SOLUTION**Q#: 10****Marks: 3**TOPIC: **Coordinate Geometry**

Jan 2022 P1H - Jan 2022 | P1H

WORKED SOLUTION*Dr. Eslam Ahmed***Topic check.** Coordinate geometry. The tag is correct.**Solution**

The line is

$$2y + 7x = 10$$

$$2y = -7x + 10$$

$$y = -\frac{7}{2}x + 5$$

So the gradient is

$$-\frac{7}{2}$$

For the x-axis, $y = 0$:

$$2(0) + 7x = 10$$

$$x = \frac{10}{7}$$

FINAL ANSWERGradient $-\frac{7}{2}$, and x-axis intersection $\left(\frac{10}{7}, 0\right)$.

PAST PAPER QUESTION**Q#: 11****Marks: 3****TOPIC: Compound Interest & Depreciation**

Jan 2022 P1H - Jan 2022 | P1H

- 11** Himari invests 200 000 yen for 3 years in a savings account paying compound interest.

The rate of interest is 1.8% for the first year and $x\%$ for each of the second year and the third year.

The value of the investment at the end of the third year is 209 754 yen.

Work out the value of x

Give your answer correct to one decimal place.

$x = \dots\dots\dots$

(Total for Question 11 is 3 marks)

PAST PAPER SOLUTION**Q#: 11****Marks: 3****TOPIC: Compound Interest & Depreciation**

Jan 2022 P1H - Jan 2022 | P1H

WORKED SOLUTION*Dr. Eslam Ahmed***Topic check.** Compound Interest and Depreciation. The tag is correct.**Method**

The first-year multiplier is

$$1.018$$

The second and third years both have multiplier $1 + \frac{x}{100}$, so

$$200000(1.018) \left(1 + \frac{x}{100}\right)^2 = 209754$$

$$\left(1 + \frac{x}{100}\right)^2 = \frac{209754}{200000(1.018)}$$

$$1 + \frac{x}{100} = 1.015000\dots$$

$$x = 1.5000\dots$$

FINAL ANSWER

$$x = 1.5$$

PAST PAPER QUESTION**Q#: 12****Marks: 6****TOPIC: Cumulative Frequency Diagrams**

Jan 2022 P1H - Jan 2022 | P1H

12 The table gives information about the times, in minutes, taken by 80 customers to do their shopping in a supermarket.

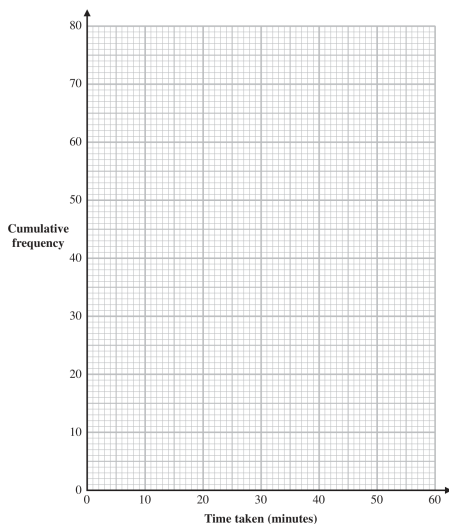
Time taken (t minutes)	Frequency
$0 < t \leq 10$	7
$10 < t \leq 20$	26
$20 < t \leq 30$	24
$30 < t \leq 40$	14
$40 < t \leq 50$	7
$50 < t \leq 60$	2

(a) Complete the cumulative frequency table.

Time taken (t minutes)	Cumulative frequency
$0 < t \leq 10$	
$0 < t \leq 20$	
$0 < t \leq 30$	
$0 < t \leq 40$	
$0 < t \leq 50$	
$0 < t \leq 60$	

(1)

(b) On the grid opposite, draw a cumulative frequency graph for your table.



(2)

(c) Use your graph to find an estimate for the median time taken.

_____ minutes

(1)

One of the 80 customers is chosen at random.

(d) Use your graph to find an estimate for the probability that the time taken by this customer was more than 42 minutes.

(2)

(Total for Question 12 is 6 marks)

PAST PAPER SOLUTION**Q#: 12****Marks: 6****TOPIC: Cumulative Frequency Diagrams**

Jan 2022 P1H - Jan 2022 | P1H

WORKED SOLUTION*Dr. Eslam Ahmed***Topic check.** Cumulative Frequency Diagrams. The tag is correct.**Solution**

Complete the cumulative frequencies:

7, 33, 57, 71, 78, 80

Plot the points

 $(0, 0), (10, 7), (20, 33), (30, 57), (40, 71), (50, 78), (60, 80)$

The median is at cumulative frequency 40.

This is between (20, 33) and (30, 57):

$$20 + \frac{40 - 33}{57 - 33} \times 10 = 22.9$$

So the median is about 23 minutes.

At 42 minutes:

$$71 + \frac{2}{10}(78 - 71) = 72.4$$

Number taking more than 42 minutes:

$$80 - 72.4 = 7.6$$

$$P(t > 42) \approx \frac{7.6}{80} = 0.095$$

FINAL ANSWER

cumulative frequencies 7, 33, 57, 71, 78, 80, median about 23 minutes, probability about 0.095.

PAST PAPER QUESTION**Q#: 13****Marks: 6****TOPIC: Expanding Brackets**

Jan 2022 P1H - Jan 2022 | P1H

13 (a) Expand and simplify $5x(x + 2)(3x - 4)$ (b) Simplify completely $\left(\frac{16w^8}{y^{20}}\right)^{-\frac{3}{4}}$
(3).....
(3)**(Total for Question 13 is 6 marks)**

PAST PAPER SOLUTION**Q#: 13****Marks: 6****TOPIC: Expanding Brackets**

Jan 2022 P1H - Jan 2022 | P1H

WORKED SOLUTION*Dr. Eslam Ahmed***Topic check.** Expanding brackets. The tag is correct.**Solution**

(a)

$$\begin{aligned}
 5x(x+2)(3x-4) &= 5x(3x^2 + 2x - 8) \\
 &= 15x^3 + 10x^2 - 40x
 \end{aligned}$$

(b)

$$\left(\frac{16w^8}{y^{20}}\right)^{-3/4} = 16^{-3/4}w^{-6}y^{15}$$

Since $16^{3/4} = 8$,

$$= \frac{y^{15}}{8w^6}$$

FINAL ANSWER(a) $15x^3 + 10x^2 - 40x$, (b) $\frac{y^{15}}{8w^6}$

PAST PAPER SOLUTION**Q#: 14****Marks: 4****TOPIC: Probability Diagrams - Venn & Tree Diagrams**

Jan 2022 P1H - Jan 2022 | P1H

WORKED SOLUTION*Dr. Eslam Ahmed***Topic check.** Probability Diagrams - Venn & Tree Diagrams. The tag is correct.**Solution**

Packet A:

$$P(\text{sunflower}) = \frac{7}{12}$$

Packet B:

$$P(\text{sunflower}) = \frac{8}{15}$$

So

$$\begin{aligned} P(\text{two sunflower seeds}) &= \frac{7}{12} \times \frac{8}{15} \\ &= \frac{14}{45} \end{aligned}$$

FINAL ANSWER

$$\frac{14}{45}$$

PAST PAPER QUESTION**Q#: 15****Marks: 3****TOPIC: Direct & Inverse Proportion**

Jan 2022 P1H - Jan 2022 | P1H

15 A is inversely proportional to C^2

$A = 40$ when $C = 1.5$

Calculate the value of C when $A = 1000$

$C = \dots\dots\dots$

(Total for Question 15 is 3 marks)

PAST PAPER SOLUTION**Q#: 15****Marks: 3****TOPIC: Direct & Inverse Proportion**

Jan 2022 P1H - Jan 2022 | P1H

WORKED SOLUTION*Dr. Eslam Ahmed***Topic check.** Direct and Inverse Proportion. The tag is correct.**Method**

A is inversely proportional to C^2 , so

$$A = \frac{k}{C^2}$$

Use $A = 40$ when $C = 1.5$:

$$40 = \frac{k}{1.5^2}$$

$$k = 40 \times 1.5^2 = 90$$

So

$$A = \frac{90}{C^2}$$

When $A = 1000$,

$$1000 = \frac{90}{C^2}$$

$$C^2 = 0.09$$

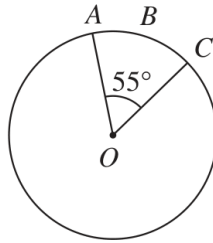
$$C = 0.3$$

FINAL ANSWER

$C = 0.3$.

PAST PAPER QUESTION**Q#: 16****Marks: 4****TOPIC: Circles, Arcs & Sectors**

Jan 2022 P1H - Jan 2022 | P1H

16 The diagram shows a circle with centre O Diagram **NOT**
accurately drawn A , B and C are points on the circle so that the length of the arc ABC is 5 cm.Given that angle $AOC = 55^\circ$

work out the area of the circle.

Give your answer correct to one decimal place.

..... cm^2 **(Total for Question 16 is 4 marks)**

PAST PAPER SOLUTION**Q#: 16****Marks: 4****TOPIC: Circles, Arcs & Sectors**

Jan 2022 P1H - Jan 2022 | P1H

WORKED SOLUTION*Dr. Eslam Ahmed***Topic check.** Circles, Arcs & Sectors. The tag is correct.**Solution**The arc length is 5 cm, and the angle is 55° .

$$5 = \frac{55}{360} \times 2\pi r$$

$$r = 5.208 \dots$$

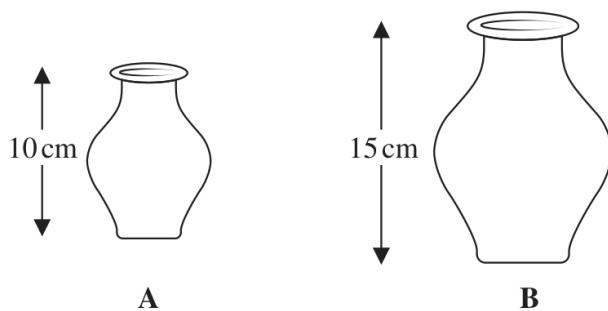
Area of the circle:

$$\pi r^2 = \pi(5.208 \dots)^2 = 85.233 \dots$$

FINAL ANSWER85.2 cm².*Dr. Eslam*

PAST PAPER QUESTION**Q#: 17****Marks: 4****TOPIC: Area & Volume of Similar Shapes**

Jan 2022 P1H - Jan 2022 | P1H

17 **A** and **B** are two similar vases.Vase **A** has height 10 cm.Vase **B** has height 15 cm.The difference between the volume of vase **A** and the volume of vase **B** is 1197 cm^3 Calculate the volume of vase **A**..... cm^3 **(Total for Question 17 is 4 marks)**

PAST PAPER SOLUTION**Q#: 17****Marks: 4****TOPIC: Area & Volume of Similar Shapes**

Jan 2022 P1H - Jan 2022 | P1H

WORKED SOLUTION*Dr. Eslam Ahmed***Topic check.** Area & Volume of Similar Shapes. The tag is correct.**Solution**The linear scale factor from vase A to vase B is

$$\frac{15}{10} = 1.5$$

So the volume scale factor is

$$1.5^3 = 3.375$$

Let the volume of vase A be V .

$$3.375V - V = 1197$$

$$2.375V = 1197$$

$$V = 504$$

FINAL ANSWER504 cm³.

PAST PAPER QUESTION**Q#: 18****Marks: 3****TOPIC: Rounding, Estimation & Bounds**

Jan 2022 P1H - Jan 2022 | P1H

18 $A = w - \frac{x^2}{y}$

$w = 3.45$ correct to 2 decimal places.

$x = 1.9$ correct to 1 decimal place.

$y = 5$ correct to the nearest whole number.

Work out the lower bound of the value of A
Show your working clearly.

(Total for Question 18 is 3 marks)

PAST PAPER SOLUTION**Q#: 18****Marks: 3****TOPIC: Rounding, Estimation & Bounds**

Jan 2022 P1H - Jan 2022 | P1H

WORKED SOLUTION*Dr. Eslam Ahmed***Topic check.** Rounding, Estimation and Bounds. The tag is correct.**Method**

$$A = w - \frac{x^2}{y}$$

For the lower bound, use the lower bound of w , and make $\frac{x^2}{y}$ as large as possible.

$$w \geq 3.445, \quad x < 1.95, \quad y \geq 4.5$$

$$A_{\text{lower}} = 3.445 - \frac{1.95^2}{4.5}$$

$$A_{\text{lower}} = 2.6$$

FINAL ANSWER

2.6

PAST PAPER QUESTION**Q#: 19****Marks: 5****TOPIC: Simultaneous Equations**

Jan 2022 P1H - Jan 2022 | P1H

19 Solve the simultaneous equations

$$\begin{aligned} 3x^2 + y^2 - xy &= 5 \\ y &= 2x - 3 \end{aligned}$$

Show clear algebraic working.

(Total for Question 19 is 5 marks)

PAST PAPER SOLUTION**Q#: 19****Marks: 5****TOPIC: Simultaneous Equations**

Jan 2022 P1H - Jan 2022 | P1H

WORKED SOLUTION*Dr. Eslam Ahmed***Topic check.** Simultaneous equations. The tag is correct.**Solution**

$$3x^2 + y^2 - xy = 5$$

$$y = 2x - 3$$

Substitute $y = 2x - 3$:

$$3x^2 + (2x - 3)^2 - x(2x - 3) = 5$$

$$3x^2 + 4x^2 - 12x + 9 - 2x^2 + 3x = 5$$

$$5x^2 - 9x + 9 = 5$$

$$5x^2 - 9x + 4 = 0$$

Factorise:

$$(5x - 4)(x - 1) = 0$$

So

$$x = \frac{4}{5} \quad \text{or} \quad x = 1$$

If $x = \frac{4}{5}$:

$$y = 2\left(\frac{4}{5}\right) - 3 = -\frac{7}{5}$$

If $x = 1$:

$$y = 2(1) - 3 = -1$$

FINAL ANSWER

$$(x, y) = \left(\frac{4}{5}, -\frac{7}{5}\right) \text{ or } (1, -1).$$

PAST PAPER QUESTION**Q#: 20****Marks: 4****TOPIC: Completing the Square**

Jan 2022 P1H - Jan 2022 | P1H

20 (a) Express $7 + 12x - 3x^2$ in the form $a + b(x + c)^2$ where a , b and c are integers.

.....
(3)

C is the curve with equation $y = 7 + 12x - 3x^2$

The point **A** is the maximum point on **C**

(b) Use your answer to part (a) to write down the coordinates of **A**

(.....,)
(1)

(Total for Question 20 is 4 marks)

PAST PAPER SOLUTION**Q#: 20****Marks: 4****TOPIC: Completing the Square**

Jan 2022 P1H - Jan 2022 | P1H

WORKED SOLUTION*Dr. Eslam Ahmed***Topic check.** Correct.**Method**

(a)

$$7 + 12x - 3x^2 = -3x^2 + 12x + 7$$

$$= -3(x^2 - 4x) + 7$$

$$= -3((x - 2)^2 - 4) + 7$$

$$= 19 - 3(x - 2)^2$$

This is the same as

$$19 - 3(x + (-2))^2$$

(b) The maximum point is read from the completed square form.

$$y = 19 - 3(x - 2)^2$$

The maximum occurs when $x = 2$, giving $y = 19$.**FINAL ANSWER**(a) $19 - 3(x - 2)^2$, (b) $A = (2, 19)$

PAST PAPER QUESTION**Q#: 20****Marks: 4****TOPIC: Completing the Square**

Jan 2022 P1H - Jan 2022 | P1H

20 (a) Express $7 + 12x - 3x^2$ in the form $a + b(x + c)^2$ where a , b and c are integers.

.....
(3)

C is the curve with equation $y = 7 + 12x - 3x^2$

The point **A** is the maximum point on **C**

(b) Use your answer to part (a) to write down the coordinates of **A**

(.....,)
(1)

(Total for Question 20 is 4 marks)

PAST PAPER SOLUTION**Q#: 20****Marks: 4****TOPIC: Completing the Square**

Jan 2022 P1H - Jan 2022 | P1H

WORKED SOLUTION*Dr. Eslam Ahmed***Topic check.** Correct.**Method**

(a)

$$7 + 12x - 3x^2 = -3x^2 + 12x + 7$$

$$= -3(x^2 - 4x) + 7$$

$$= -3((x - 2)^2 - 4) + 7$$

$$= 19 - 3(x - 2)^2$$

This is the same as

$$19 - 3(x + (-2))^2$$

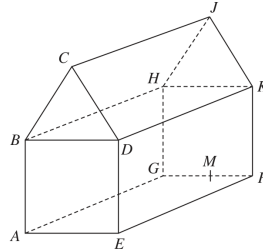
(b) The maximum point is read from the completed square form.

$$y = 19 - 3(x - 2)^2$$

The maximum occurs when $x = 2$, giving $y = 19$.**FINAL ANSWER**(a) $19 - 3(x - 2)^2$, (b) $A = (2, 19)$

PAST PAPER QUESTION**Q#: 21****Marks: 5****TOPIC: 3D Pythagoras & Trigonometry**

Jan 2022 P1H - Jan 2022 | P1H

21 The diagram shows the prism $ABCDEFGHJK$ with horizontal base $AEFG$ Diagram **NOT**
accurately drawn

$ABCDE$ is a cross section of the prism where
 $ABDE$ is a square
 BCD is an equilateral triangle

$$EF = 2 \times AE$$

M is the midpoint of GF so that JM is vertical.

$$\text{Angle } MAJ = y^\circ$$

$$\text{Given that } \tan y^\circ = T$$

find the value of T , giving your answer in the form $\frac{\sqrt{p} + \sqrt{q}}{17}$ where p and q are integers.

$$T = \dots\dots\dots$$

(Total for Question 21 is 5 marks)

PAST PAPER SOLUTION**Q#: 21****Marks: 5****TOPIC: 3D Pythagoras & Trigonometry**

Jan 2022 P1H - Jan 2022 | P1H

WORKED SOLUTION

Dr. Eslam Ahmed

Topic check. Moved from Coordinate Geometry to 3D Pythagoras & Trigonometry.**Method**

Let

$$AE = x$$

Since $ABDE$ is a square,

$$AB = BD = DE = AE = x$$

Since BCD is an equilateral triangle, the height from C to BD is

$$\frac{\sqrt{3}}{2}x$$

Therefore the vertical height from the base to J is

$$JM = x + \frac{\sqrt{3}}{2}x$$

$$JM = \frac{x(2 + \sqrt{3})}{2}$$

Also,

$$EF = 2AE = 2x$$

Since M is the midpoint of GF , the horizontal distance from A to M is

$$AM = \sqrt{\left(\frac{x}{2}\right)^2 + (2x)^2}$$

$$AM = \sqrt{\frac{x^2}{4} + 4x^2}$$

$$AM = \frac{x\sqrt{17}}{2}$$

In right triangle AMJ ,

$$\tan y^\circ = \frac{JM}{AM}$$

$$T = \frac{\frac{x(2+\sqrt{3})}{2}}{\frac{x\sqrt{17}}{2}}$$

$$T = \frac{2 + \sqrt{3}}{\sqrt{17}}$$

Rationalise the denominator:

$$T = \frac{(2 + \sqrt{3})\sqrt{17}}{17}$$

$$T = \frac{2\sqrt{17} + \sqrt{51}}{17}$$

$$T = \frac{\sqrt{68} + \sqrt{51}}{17}$$

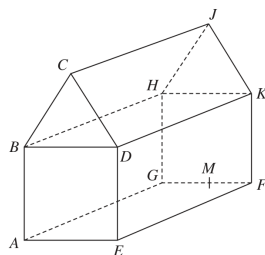
FINAL ANSWER

$$T = \frac{\sqrt{68} + \sqrt{51}}{17}$$

Dr. Eslam Ahmed

PAST PAPER QUESTION**Q#: 21** **Marks: 5****TOPIC: 3D Pythagoras & Trigonometry**

Jan 2022 P1H - Jan 2022 | P1H

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 BCD is an equilateral triangle

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 M is the midpoint of GF so that JM is vertical.

Angle $MAJ = y^\circ$

Given that $\tan y^\circ = T$

find the value of T , giving your answer in the form $\frac{\sqrt{p} + \sqrt{q}}{17}$ where p and q are integers.

$T = \dots\dots\dots$

(Total for Question 21 is 5 marks)

PAST PAPER SOLUTION**Q#: 21****Marks: 5****TOPIC: 3D Pythagoras & Trigonometry**

Jan 2022 P1H - Jan 2022 | P1H

WORKED SOLUTION*Dr. Eslam Ahmed***Topic check.** Moved from Coordinate Geometry to 3D Pythagoras & Trigonometry.**Method**

Let

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FINAL ANSWER

$$T = \frac{\sqrt{68} + \sqrt{51}}{17}$$

Dr. Eslam Ahmed

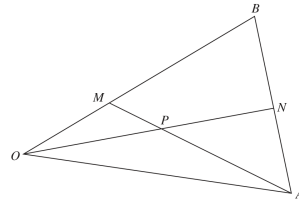
PAST PAPER QUESTION

Q#: 22

Marks: 5

TOPIC: **Vectors**

Jan 2022 P1H - Jan 2022 | P1H

22 The diagram shows triangle OAB Diagram **NOT**
accurately drawn

$$\vec{OA} = 8\mathbf{a} \quad \vec{OB} = 6\mathbf{b}$$

 M is the point on OB such that $OM:MB = 1:2$ N is the midpoint of AB P is the point of intersection of ON and AM Using a vector method, find $\vec{OP}$ as a simplified expression in terms of $\mathbf{a}$ and $\mathbf{b}$
Show your working clearly.

$$\vec{OP} = \underline{\hspace{2cm}}$$

(Total for Question 22 is 5 marks)

PAST PAPER SOLUTION**Q#: 22****Marks: 5**TOPIC: **Vectors**

Jan 2022 P1H - Jan 2022 | P1H

WORKED SOLUTION*Dr. Eslam Ahmed***Topic check.** Moved from Coordinate Geometry to Vectors.**Method**

$$\vec{OA} = 8\mathbf{a}, \quad \vec{OB} = 6\mathbf{b}$$

Since $OM : MB = 1 : 2$,

$$\vec{OM} = \frac{1}{3}\vec{OB}$$

$$\vec{OM} = 2\mathbf{b}$$

Since N is the midpoint of AB ,

$$\vec{ON} = \frac{8\mathbf{a} + 6\mathbf{b}}{2}$$

$$\vec{ON} = 4\mathbf{a} + 3\mathbf{b}$$

Point P lies on ON , so

$$\vec{OP} = \lambda(4\mathbf{a} + 3\mathbf{b})$$

Point P also lies on AM , so

$$\vec{OP} = 8\mathbf{a} + \mu(2\mathbf{b} - 8\mathbf{a})$$

$$\vec{OP} = (8 - 8\mu)\mathbf{a} + 2\mu\mathbf{b}$$

Equate coefficients:

$$4\lambda = 8 - 8\mu$$

$$3\lambda = 2\mu$$

From $3\lambda = 2\mu$,

$$\mu = \frac{3}{2}\lambda$$

Substitute:

$$4\lambda = 8 - 8\left(\frac{3}{2}\lambda\right)$$

$$4\lambda = 8 - 12\lambda$$

$$16\lambda = 8$$

$$\lambda = \frac{1}{2}$$

Therefore

$$\vec{OP} = \frac{1}{2}(4\mathbf{a} + 3\mathbf{b})$$

$$\vec{OP} = 2\mathbf{a} + \frac{3}{2}\mathbf{b}$$

FINAL ANSWER

$$\vec{OP} = 2\mathbf{a} + \frac{3}{2}\mathbf{b}$$

Dr. Eslam Ahmed

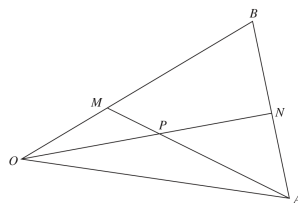
PAST PAPER QUESTION

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Marks: 5

TOPIC: **Vectors**

Jan 2022 P1H - Jan 2022 | P1H

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Show your working clearly.

$$\vec{OP} = \underline{\hspace{2cm}}$$

(Total for Question 22 is 5 marks)

PAST PAPER SOLUTION**Q#: 22****Marks: 5**TOPIC: **Vectors**

Jan 2022 P1H - Jan 2022 | P1H

WORKED SOLUTION*Dr. Eslam Ahmed***Topic check.** Moved from Coordinate Geometry to Vectors.**Method**

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$$\vec{ON} = 4\mathbf{a} + 3\mathbf{b}$$

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Equate coefficients:

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$$4\lambda = 8 - 12\lambda$$

$$16\lambda = 8$$

$$\lambda = \frac{1}{2}$$

Therefore

$$\vec{OP} = \frac{1}{2}(4\mathbf{a} + 3\mathbf{b})$$

$$\vec{OP} = 2\mathbf{a} + \frac{3}{2}\mathbf{b}$$

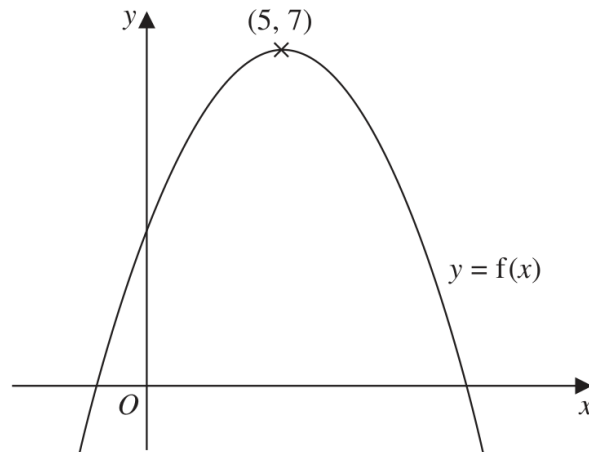
FINAL ANSWER

$$\vec{OP} = 2\mathbf{a} + \frac{3}{2}\mathbf{b}$$

Dr. Eslam Ahmed

PAST PAPER QUESTION**Q#: 23****Marks: 2****TOPIC: Transformations of Graphs**

Jan 2022 P1H - Jan 2022 | P1H

23 The diagram shows a sketch of the curve with equation $y = f(x)$ 

There is only one maximum point on the curve.
 The coordinates of this maximum point are $(5, 7)$

Write down the coordinates of the maximum point on the curve with equation

(i) $y = f(x + 9)$

(.....,)

(ii) $y = f(x) + 3$

(.....,)

(Total for Question 23 is 2 marks)

PAST PAPER SOLUTION**Q#: 23****Marks: 2****TOPIC: Transformations of Graphs**

Jan 2022 P1H - Jan 2022 | P1H

WORKED SOLUTION*Dr. Eslam Ahmed***Topic check.** Correct.**Method**The maximum point of $y = f(x)$ is $(5, 7)$.

For

$$y = f(x + 9)$$

the graph moves 9 units left:

$$(5, 7) \rightarrow (-4, 7)$$

For

$$y = f(x) + 3$$

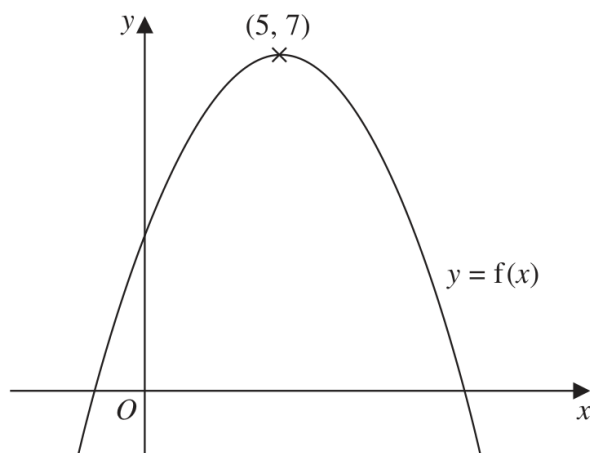
the graph moves 3 units up:

$$(5, 7) \rightarrow (5, 10)$$

FINAL ANSWER $(-4, 7)$ and $(5, 10)$

PAST PAPER QUESTION**Q#: 23****Marks: 2****TOPIC: Transformations of Graphs**

Jan 2022 P1H - Jan 2022 | P1H

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Write down the coordinates of the maximum point on the curve with equation

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(.....,)

(Total for Question 23 is 2 marks)

PAST PAPER SOLUTION**Q#: 23****Marks: 2****TOPIC: Transformations of Graphs**

Jan 2022 P1H - Jan 2022 | P1H

WORKED SOLUTION*Dr. Eslam Ahmed***Topic check.** Correct.**Method**The maximum point of $y = f(x)$ is $(5, 7)$.

For

$$y = f(x + 9)$$

the graph moves 9 units left:

$$(5, 7) \rightarrow (-4, 7)$$

For

$$y = f(x) + 3$$

the graph moves 3 units up:

$$(5, 7) \rightarrow (5, 10)$$

FINAL ANSWER $(-4, 7)$ and $(5, 10)$

PAST PAPER QUESTION**Q#: 24****Marks: 6****TOPIC: Differentiation**

Jan 2022 P1H - Jan 2022 | P1H

24 The curve **C** has equation $y = ax^3 + bx^2 - 12x + 6$ where a and b are constants.

The point A with coordinates $(2, -6)$ lies on **C**

The gradient of the curve at A is 16

Find the y coordinate of the point on the curve whose x coordinate is 3

Show clear algebraic working.

$y = \dots\dots\dots$

(Total for Question 24 is 6 marks)

PAST PAPER SOLUTION**Q#: 24****Marks: 6****TOPIC: Differentiation**

Jan 2022 P1H - Jan 2022 | P1H

WORKED SOLUTION*Dr. Eslam Ahmed***Topic check.** Correct.**Method**

$$y = ax^3 + bx^2 - 12x + 6$$

Point $A = (2, -6)$ lies on the curve:

$$-6 = 8a + 4b - 24 + 6$$

$$8a + 4b = 12$$

$$2a + b = 3$$

Differentiate:

$$\frac{dy}{dx} = 3ax^2 + 2bx - 12$$

The gradient at $x = 2$ is 16:

$$16 = 12a + 4b - 12$$

$$12a + 4b = 28$$

$$3a + b = 7$$

Now solve

$$2a + b = 3$$

$$3a + b = 7$$

Subtracting gives

$$a = 4$$

Then

$$2(4) + b = 3$$

$$b = -5$$

For $x = 3$,

$$y = 4(3)^3 - 5(3)^2 - 12(3) + 6$$

$$y = 108 - 45 - 36 + 6$$

$$y = 33$$

FINAL ANSWER

$$y = 33$$

Dr. Eslam Ahmed

PAST PAPER QUESTION**Q#: 24****Marks: 6****TOPIC: Differentiation**

Jan 2022 P1H - Jan 2022 | P1H

24 The curve **C** has equation $y = ax^3 + bx^2 - 12x + 6$ where a and b are constants.

The point A with coordinates $(2, -6)$ lies on **C**

The gradient of the curve at A is 16

Find the y coordinate of the point on the curve whose x coordinate is 3

Show clear algebraic working.

$y = \dots\dots\dots$

(Total for Question 24 is 6 marks)

PAST PAPER SOLUTION**Q#: 24****Marks: 6**TOPIC: **Differentiation**

Jan 2022 P1H - Jan 2022 | P1H

WORKED SOLUTION*Dr. Eslam Ahmed***Topic check.** Correct.**Method**

$$y = ax^3 + bx^2 - 12x + 6$$

Point $A = (2, -6)$ lies on the curve:

$$-6 = 8a + 4b - 24 + 6$$

$$8a + 4b = 12$$

$$2a + b = 3$$

Differentiate:

$$\frac{dy}{dx} = 3ax^2 + 2bx - 12$$

The gradient at $x = 2$ is 16:

$$16 = 12a + 4b - 12$$

$$12a + 4b = 28$$

$$3a + b = 7$$

Now solve

$$2a + b = 3$$

$$3a + b = 7$$

Subtracting gives

$$a = 4$$

Then

$$2(4) + b = 3$$

$$b = -5$$

For $x = 3$,

$$y = 4(3)^3 - 5(3)^2 - 12(3) + 6$$

$$y = 108 - 45 - 36 + 6$$

$$y = 33$$

FINAL ANSWER

$$y = 33$$

Dr. Eslam Ahmed